CiteWise: Evidence Retrieval in Scientific Papers

Phase 9 final validation report | Mini-project report
QASPER official splits | Paragraph-level retrieval | 25 October 2026

1. Introduction

Researchers looking for evidence in a paper must find passages that support a question, often across long sections of technical text. CiteWise addresses this retrieval step: a user selects a real QASPER paper, asks a question, and receives ranked paragraph candidates with section headings and stable evidence IDs. It does not generate answers or certify scientific truth.

The study asks: *How do lexical retrieval, a QASPER-trained lexical ranker, and four pretrained neural encoders differ in evidence ranking and CPU cost?* The contribution is a controlled comparison on the official paper-level split, a documented supervised-ranker failure, a reproducible pipeline, and an application that returns passages from the source paper.

Phase 9 verifies the saved six-method QASPER validation comparison: BM25 and supervised logistic baselines, plus MiniLM-L6-v2, BGE-base-en-v1.5, E5-base-v2 and GTE-base. These are the four faculty-specified neural retrievers. This within-paper comparison does not claim the checkpoints are the newest model families or reproduce published leaderboards.

The held-out test evaluation was performed once before E5 and GTE were added, using the original four methods. Phase 9 used validation artifacts only and did not rerun or tune against test; E5/GTE have no test scores. CiteWise remains locally tested; no public deployment or manual assistive-technology evaluation is claimed.

2. Literature Review

QASPER frames information seeking around a particular research paper. Dasigi et al. collected questions from readers who saw a paper title and abstract, then gathered answers and supporting evidence from full text [1]. This provides human evidence paragraphs for retrieval evaluation. CiteWise uses the paper-local setting and mapped textual evidence, while excluding unanswerable and unmatched questions from ranking metrics.

Chen et al.'s M3-Embedding combines dense, sparse, and multi-vector retrieval and supports multilingual inputs and long documents [2]. Its multi-function training is relevant to retrieval versatility, but CiteWise neither uses nor reproduces M3-Embedding. BGE-M3 was an intended model comparison and has no measured CiteWise score.

Saad-Falcon et al.'s ARES evaluates retrieval-augmented generation along context relevance, answer faithfulness, and answer relevance, using trained language-model judges and a small human-labeled set for prediction-powered inference [3]. CiteWise instead evaluates paragraph ranking against QASPER annotations. ARES motivates keeping retrieval quality distinct from generated-answer quality; CiteWise does not claim answer-faithfulness metrics.

Zhang et al.'s Qwen3 Embedding preprint describes embedding and reranking checkpoints at multiple scales, trained with large-scale pretraining and supervised fine-tuning [4]. It motivates comparing quality with resource cost. Qwen3-Embedding-0.6B was not downloaded or evaluated here, so no published benchmark number is attributed to CiteWise.

2. Literature Review (continued)

Wang et al.'s Astute RAG studies imperfect retrieved context and conflicts between parametric and retrieved knowledge, then proposes a source-aware generation process [5]. CiteWise shares the concern that a plausible passage can still be misleading, but stops at ranking and displays source text for inspection. It does not implement Astute RAG's generation or conflict-resolution pipeline.

Liu and Zhang's Exp4Fuse applies LLM-based query expansion to sparse retrieval, forms rankings for original and expanded queries, and fuses the lists [6]. This supports studying rank fusion as a future extension. CiteWise compares the current retrievers without query expansion and makes no Exp4Fuse implementation or performance claim.

Asai et al.'s OpenScholar indexes a large scholarly corpus and synthesizes citation-backed responses to multi-paper scientific questions [7]. It demonstrates the broader research-synthesis setting. CiteWise is narrower: retrieval is constrained to one paper and evaluated against annotated paragraphs.

Together these works distinguish retrieval from answer generation, and scientific search from multi-paper synthesis. CiteWise contributes a reproducible six-method comparison of paragraph evidence ranking on QASPER. Its findings are specific to this dataset, split, preprocessing and CPU setup; they are not reproductions of the papers' published benchmarks.

3. Methodology

Architecture. A QASPER paper is loaded with original paragraph IDs, section names and annotations. The user selects a paper and question. A selected retriever scores every original paragraph in that paper; the API returns ranked passage IDs, text, headings, scores and latency. The UI labels them evidence candidates, not verified answers.

Data discipline. The logistic classifier was fitted on train papers only (`max_iter=1500`, `random_state=42`). Model comparisons and decisions use the fixed validation split. The held-out test result remains a separate earlier four-method experiment; no Phase 8 tuning or scoring used test questions.

Neural encoders. MiniLM is the compact six-layer encoder and uses plain text, mean pooling and 256-token truncation. BGE-base uses a retrieval instruction on queries, no passage instruction, CLS pooling, normalized embeddings and a 512-token limit. E5 uses asymmetric `query:` and `passage:` prefixes, masked mean pooling and normalized 512-token embeddings. GTE is a general text-embedding model trained on relevance pairs; CiteWise uses plain text, mean pooling and a 512-token limit. All four rank normalized float32 embeddings by dot product. Different training objectives, input formatting and truncation can yield different rankings; this comparison does not isolate one cause. Immutable model revisions and evaluation checkpoint identifiers are listed on page 8.

Metrics. For gold paragraph set G, Recall@k = |G intersect top-k| / |G|. Recall@1 was calculated from the first saved validation candidate. MRR@5 is the reciprocal rank of the first relevant passage or zero. Binary nDCG@5 discounts relevant results by log rank and divides by ideal gain. Metrics are macro-averaged over scorable questions. Percentile 95% intervals use 2,000 paper-cluster bootstrap replicates with seed 42; paired comparisons use identical questions.

Timing and checkpoint provenance. Warm P50/P95 includes query encoding, scoring and sorting with model/document vectors cached. Model loading and paragraph encoding are timed separately. A Windows checkpoint-write repair was applied mid-run; retrieval/evaluation code hashes, data SHA, query-plan SHA and protocol match across runs. Per-model source hashes and the checkpoint-writer migration are retained in `artifacts/phase8/full-validation/benchmark_validation.json`.

4. Data Cleaning and Preprocessing

The source is Allen Institute's QASPER dataset, revision `91bc63e221475d59a6e17e4299fbfc5a919d7c7a`, distributed under CC BY 4.0 [1]. The official paper-level partitions are retained; no random paragraph split or cross-split resampling was performed. The prepared data audit records:

Official split	Papers	Non-empty paragraphs	Questions	Scorable	Unmatched/no text	Unanswerable
Train	888	46,882	2,593	2,098	224	271
Validation	281	13,266	1,005	888	57	60
Test	416	19,817	1,451	1,309	63	79

Preparation strips empty paragraphs and collapses whitespace. Evidence phrases are whitespace-normalized and case-folded before mapping to stable IDs tied to original section/paragraph positions. Repeated identical paragraph text can map to multiple IDs. Figure/table-only annotations are counted as unsupported textual evidence rather than converted into fabricated paragraphs: 386 items in train, 253 in validation, and 459 in test.

A question is scorable if at least one annotated textual evidence phrase maps to a paragraph. Unanswerable questions and questions without mapped text evidence are excluded consistently for every method and reported as coverage exclusions. The final test run evaluated all 1,309 scorable questions across 408 papers; its 63 unmatched/no-text and 79 unanswerable questions were excluded. The final test questions were not used for training or model selection.

5. Results and Discussion

The complete six-method comparison evaluated all **888 scorable validation questions** across **276 papers**. Every model has the same query, source-paper paragraphs and mapped gold evidence. The intervals are 95% paper-cluster bootstrap estimates (2,000 replicates, seed 42).

Method	Recall@1 [95% CI]	Recall@5 [95% CI]	MRR@5 [95% CI]	nDCG@5 [95% CI]
BM25	13.96% [11.95, 16.07]	44.32% [41.04, 47.44]	34.56% [32.01, 37.21]	32.86% [30.33, 35.32]
Supervised logistic	0.71% [0.35, 1.16]	4.93% [3.74, 6.27]	5.20% [4.13, 6.38]	3.91% [3.09, 4.86]
MiniLM-L6-v2	14.04% [11.89, 16.31]	46.11% [43.14, 48.99]	36.05% [33.34, 38.91]	34.57% [32.20, 36.95]
BGE-base-en-v1.5	20.11% [17.87, 22.52]	52.10% [49.20, 55.08]	43.87% [40.99, 46.60]	41.42% [38.98, 43.95]
E5-base-v2	21.66% [19.10, 24.16]	54.67% [51.54, 57.63]	46.09% [43.27, 48.89]	43.40% [40.83, 45.79]
GTE-base	13.93% [11.83, 16.14]	45.69% [42.65, 48.69]	36.19% [33.27, 38.89]	34.51% [32.05, 37.00]

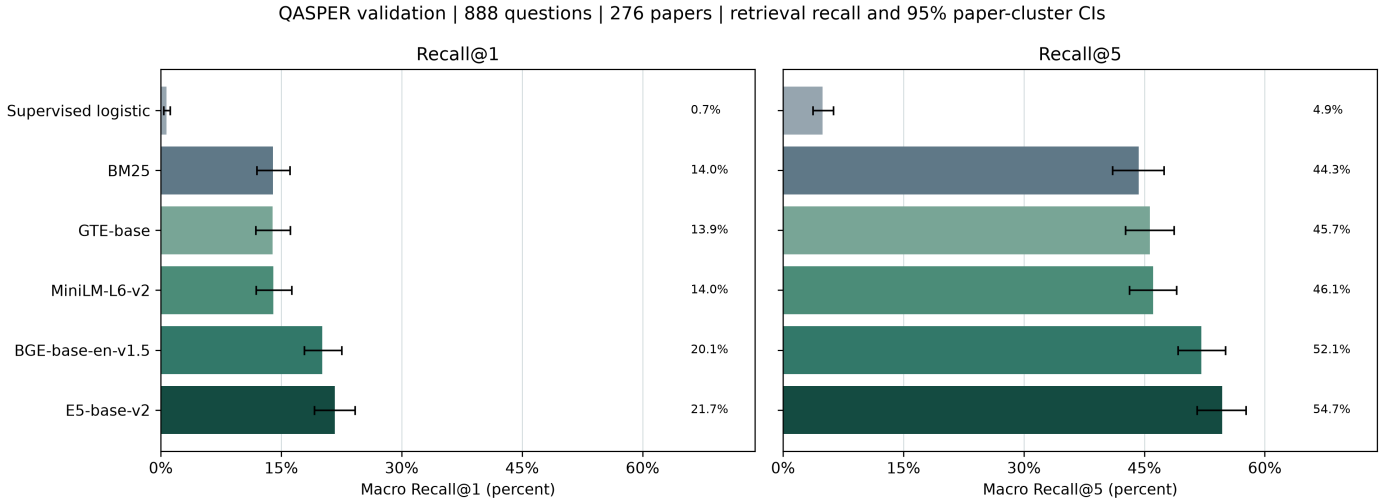


Figure 1. Macro Recall@1 and Recall@5 on the same 888 scorable validation questions. Whiskers are 95% paper-cluster bootstrap intervals; E5-base-v2 has the highest neural point estimate for both.

5. Results and Discussion (continued)

Paired bootstrap comparisons use the same queries and resample 276 papers. E5-minus-MiniLM and E5-minus-GTE 95% intervals exclude zero on all four metrics. Against BGE, E5 differences are Recall@1 +1.55 pp [-0.78, 4.01], Recall@5 +2.57 [0.04, 5.26], MRR +2.21 [-0.17, 4.60], and nDCG +1.98 [-0.07, 4.06]. Only the E5-BGE Recall@5 interval excludes zero. Intervals are unadjusted for multiple comparisons.

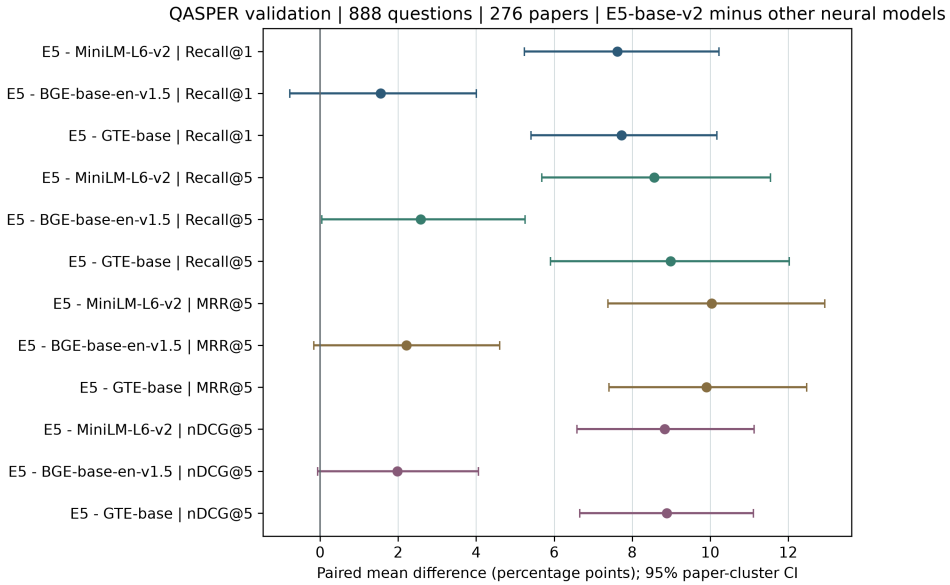


Figure 2. E5-base-v2 minus each other neural retriever, derived from saved per-query rankings; points are paired mean differences and whiskers are 95% paper-cluster intervals.

Method	Warm query P50/P95 (ms)	Model load (ms)	Paragraph encoding (s)	Peak worker RSS (GiB)
BM25	2.70 / 5.26	0.01	0.0	0.06
Supervised logistic	4.81 / 9.31	4,809.5	0.0	0.18
MiniLM-L6-v2	18.81 / 26.22	659.8	421.9	0.70
BGE-base-en-v1.5	88.18 / 110.41	1,081.2	3,029.0	1.18
E5-base-v2	73.31 / 91.23	1,042.1	2,982.8	1.16
GTE-base	70.83 / 88.80	999.3	2,958.6	1.17

CPU-only Windows 11 measurements used 16 logical CPUs and 15.82 GiB RAM, with no CUDA. Document encoding covered 12,869 paragraphs; warm query timing excludes loading and encoding and is not a concurrent-service estimate. The train diagnostic sampled 3,603 positives and 12,581 negatives (22.3% positive); mean BM25 was 3.329/5.044 for sampled positives/hard negatives but 3.055/1.704 for full-validation positives/non-relevant candidates. Standardization left the bounded validation Recall/MRR/nDCG at 0.0175/0.0333/0.0182. This supports candidate-distribution mismatch, not a single-cause explanation.

The raw-token audit counted 569 MiniLM passages above 256 tokens, 21 BGE, 22 E5 and 21 GTE passages above 512. The `273 > 256` warning is from untruncated auditing; inference truncates MiniLM at 256. All models completed 888 questions without indexing errors.

The table below preserves the **earlier, one-time held-out test** results for the original four methods only. It is not a six-model comparison; E5/GTE were not tested, and Phase 8 decisions used validation only.

Frozen test method	Recall@5 (95% CI)	MRR@5 (95% CI)	nDCG@5 (95% CI)
BM25	43.77% [41.38, 46.24]	37.92% [35.63, 40.19]	34.57% [32.61, 36.58]
Supervised logistic	5.50% [4.53, 6.60]	6.66% [5.54, 7.91]	4.87% [4.03, 5.85]
MiniLM-L6-v2	47.36% [44.92, 49.87]	43.70% [41.34, 46.07]	38.77% [36.66, 40.90]
BGE-base-en-v1.5	52.35% [49.84, 54.86]	49.97% [47.45, 52.30]	44.32% [42.18, 46.47]

6. State-of-the-Art Models Comparison

The four requested neural encoders were evaluated with the same full-validation question set, paragraph universe, relevance mapping and ranking pipeline. Their quality/latency trade-off is shown below; baselines remain separate on page 6.

Neural model	Checkpoint ID prefix	Recall@1	Recall@5	MRR@5	nDCG@5	Warm P50/P95 (ms)
MiniLM-L6-v2	`2c0033877978`	14.04%	46.11%	36.05%	34.57%	18.81 / 26.22
BGE-base-en-v1.5	`1276b46e3d7c`	20.11%	52.10%	43.87%	41.42%	88.18 / 110.41
E5-base-v2	`9c213a788043`	21.66%	54.67%	46.09%	43.40%	73.31 / 91.23
GTE-base	`413f64fb5b7f`	13.93%	45.69%	36.19%	34.51%	70.83 / 88.80

Pinned model revisions are MiniLM `1110a243fdf4706b3f48f1d95db1a4f5529b4d41`, BGE `a5beb1e3e68b9ab74eb54cfd186867f64f240e1a`, E5 `1c644c92ad3ba1efdad3f1451a637716616a20e8` and GTE `f846641ce4fb595a946b68fe18389798a8da9a64`. The complete 64-character configuration checkpoint IDs and baseline IDs are in `artifacts/phase9/model_comparison.csv` and the dashboard bundle.

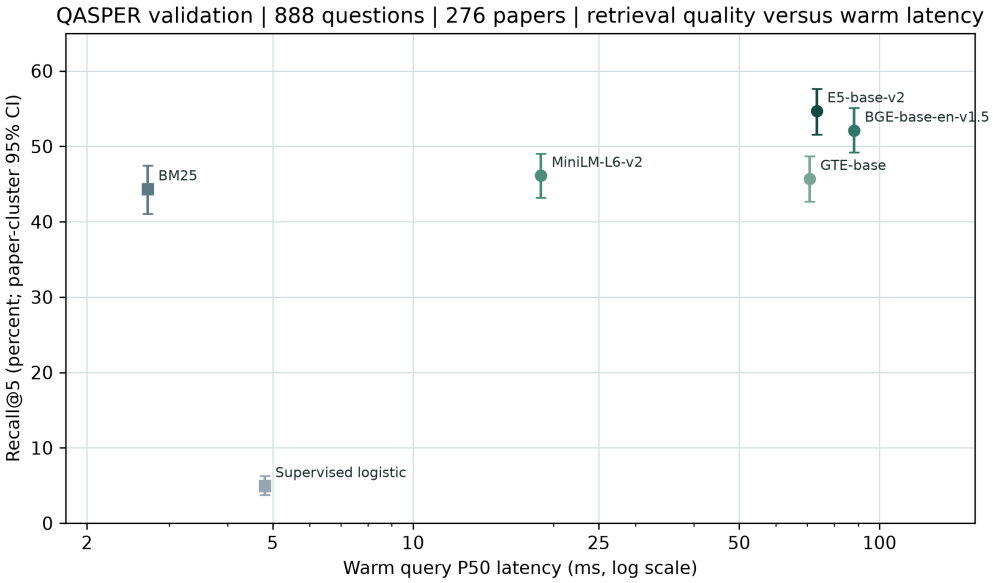


Figure 3. Recall@5 versus cached warm P50 retrieval latency. Error bars show paper-cluster 95% intervals; the x-axis is logarithmic. E5 is the validation point-estimate leader, while MiniLM is the fastest neural retriever. The separate model-loading and document-encoding chart is retained in the Phase 9 figures.

E5/GTE are prior-generation encoders relative to current embedding releases; they satisfy the project-specified four-model experiment but should not be described as the newest available systems. Published leaderboard scores are not CiteWise results.

7. User Interface Development and Deployment

CiteWise integrates FastAPI, real QASPER paper selection, model availability checks, the retrieval API and a validation-only results dashboard. Search returns actual ranked passages with section headings, IDs, source text and scores. Unavailable models are disabled; the held-out test split is rejected by the interactive API.



Figure 4. Current local desktop comparison dashboard showing the six-method validation table, paired results and method information.

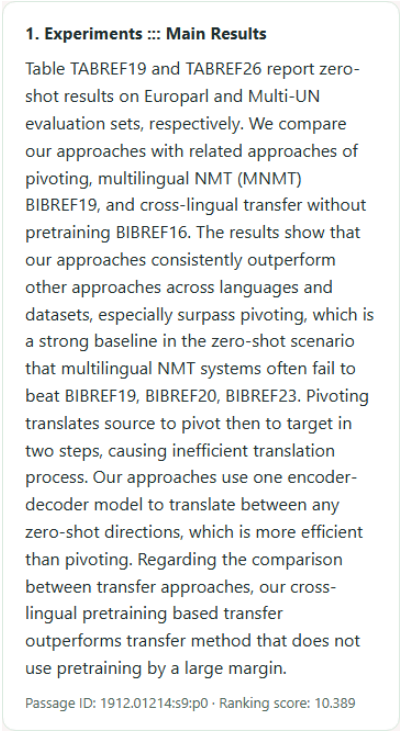


Figure 5. Source passage and evidence ID at 390 CSS pixels; 375-pixel layout also passed.

Ten Playwright browser tests passed locally, covering BM25 and all four neural models against real validation paper text, dashboard provenance, invalid requests, keyboard focus, validation-only access and 375/390-pixel layouts. Axe reported zero WCAG 2.1 A/AA violations. These are automated checks; manual screen-reader and assistive-technology evaluations were not performed.

The Dockerfile, pinned model staging and health check remain deployment instructions. Docker Engine was unavailable, so no container build was verified. There is no public HTTPS URL or hosted search flow; CiteWise is locally demonstrated, not publicly deployed.

8. References

[1] P. Dasigi, K. Lo, I. Beltagy, A. Cohan, N. A. Smith, and M. Gardner. "A Dataset of Information-Seeking Questions and Answers Anchored in Research Papers." *Proceedings of NAACL-HLT*, 2021, pp. 4599-4610. https://aclanthology.org/2021.naacl-main.365/

[2] J. Chen et al. "M3-Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through Self-Knowledge Distillation." *Findings of ACL*, 2024, pp. 2318-2335. https://aclanthology.org/2024.findings-acl.137/

[3] J. Saad-Falcon, O. Khattab, C. Potts, and M. Zaharia. "ARES: An Automated Evaluation Framework for Retrieval-Augmented Generation Systems." *NAACL-HLT*, 2024, pp. 338-354. https://aclanthology.org/2024.naacl-long.20/

[4] Y. Zhang et al. "Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models." arXiv:2506.05176, 2025. https://arxiv.org/abs/2506.05176

[5] F. Wang et al. "Astute RAG: Overcoming Imperfect Retrieval Augmentation and Knowledge Conflicts for Large Language Models." *ACL*, 2025, pp. 30553-30571. https://aclanthology.org/2025.acl-long.1476/

[6] L. Liu and M. Zhang. "Exp4Fuse: A Rank Fusion Framework for Enhanced Sparse Retrieval using Large Language Model-based Query Expansion." *Findings of ACL*, 2025, pp. 163-173. https://aclanthology.org/2025.findings-acl.9/

[7] A. Asai et al. "Synthesizing scientific literature with retrieval-augmented language models." *Nature* 650, 857-863 (2026). https://doi.org/10.1038/s41586-025-10072-4

Data and model cards. QASPER dataset card: https://huggingface.co/datasets/allenai/qasper. MiniLM: https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2. BGE: https://huggingface.co/BAAI/bge-base-en-v1.5. E5: https://huggingface.co/intfloat/e5-base-v2. GTE: https://huggingface.co/thenlper/gte-base. QASPER is CC BY 4.0; retain source attribution. Model licenses and project code-license status are in `docs/LICENSING.md`.